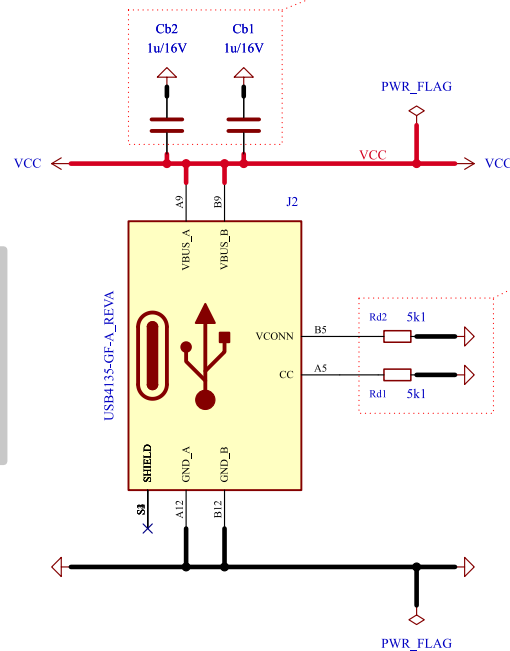


[4] USB connector



DESIGN NOTE:
Cb1, Cb2
bulk capacitor

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Table 4-2 VBUS Source Characteristics

	Minimum	Maximum	Notes
VBUS Leakage Impedance	72.4 k Ω		Leakage between VBUS pins and GND pins on receptacle when VBUS is not being sourced.
VBUS Capacitance		3000 μ F	Capacitance for source-only ports between VBUS and GND pins on receptacle when VBUS is not being sourced.
		10 μ F	Capacitance for DRP ports between VBUS and GND pins on receptacle when VBUS is not being sourced.

DESIGN NOTE:

Per USB-IF specification, a sink role is established by placing 5.1 k Ω pull-down (Rd) resistors on CC1 and CC2.

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Table 4-25 Sink CC Termination (Rd) Requirements

Rd Implementation	Nominal value	Can detect power capability?	Max voltage on pin
$\pm 20\%$ voltage clamp ¹	1.1 V	No	1.32 V
$\pm 20\%$ resistor to GND	5.1 k Ω	No	2.18 V
$\pm 10\%$ resistor to GND	5.1 k Ω	Yes	2.04 V

Notes:

1. The clamp implementation inhibits *USB PD* communication although the system can start with the clamp and transition to the resistor once it is able to do *USB PD*.

www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf

Comments:

AN5346
STM32G474 Datasheet p.81
J. Pieper ADC investigation

Company:

BaJRan imgaginary industries

Variant:

CHECKED

Git Hash:

5ba43f3

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

USB connector

File Name:

Section A - USB connector.kicad_sch

Designer:

Pawel Baran

Date:

2025-01-12

Revision:

2.1.1+ (Unreleased)

Sheet Path:

/Project Architecture/Section A - USB connector/

Reviewer:

Size:

A4

Sheet:

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